AI ETHICS

AI for Social Impact in the Global South

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This lecture describes AI work carried out for social impact (AI4SI) in the developing world, at the Wadhwani Institute for Artificial Intelligence in India. We introduce what we mean by AI4SI, briefly describe how this work ties in to the UN Sustainable Development Goals, and the overall challenges in introducing and deploying AI technology for social upliftment in areas where technology adoption is limited, and AI is poorly understood. Finally, we describe three case studies of specific AI projects we are working on in the agriculture and healthcare domains, as well as challenges encountered in these projects at various stages of development, from data to adoption to deployment.

Lecture Transcript

0:03 Hello, everyone. I'm Alpan Raval. And this lecture is on AI for Social Impact in the Global South. I am Chief Scientist at Wadhwani AI, which is a nonprofit institute located in India, whose charter is to build and deploy AI solutions for use by underserved populations across the Global South. I'm a theoretical physicist by training, but I've been working in the applied AI space, on and off for a couple of decades now. Our work because of really, our customer base is relevant to the global discussion around AI and ethics, especially as it relates to using AI for the betterment of lives and livelihoods in communities that do not have much exposure to technology beyond a very basic smartphone. I think we bring a somewhat unique perspective to this topic, because of the types of challenges we face around things like data issues and deployment on the ground.

1:05 So the outline of this lecture is as follows. I first, say a little bit about what we mean by AI for Social Impact, and how it links with the UN SDGs. I'll describe briefly our work and overall approach. I'll describe the high level challenges in this space. And then really get into three different solutions that we are working on, among many other things, that highlight some of these challenges, highlight how solution development takes place. And these three areas are: automated pest management in cotton farms in India, AI powered newborn anthropometry and AI-based prediction of loss to follow up risk in Tuberculosis patients.

1:57 So just to give some context behind the work we do, and especially as it stands in relation to this course, we know that modern societal problems are complex. They affect lives and livelihoods across the developing world, especially, in a very dynamic and interconnected way. AI has a natural ability as a technology to use disparate data to amplify the reach of decision making systems and to uncover some of these complexities. And therefore, it's a technology that's well suited to address these types of problems.

2:37 Now even though the potential for good with AI is huge, deployment, adoption and impact of AI models in this space, you know, societal problems affecting the developing world, that's the space I mean, in this space is very limited. And the idea behind AI for Social Impact is to address this gap. In a way, we see this as complementary or perhaps even a subset of AI for Social Good, which nowadays has increasingly come to mean the resolution of negative effects of large scale AI, especially as they relate to things like fairness, bias and inequity, which are all important problems. But large scale AI is mostly deployed in affluent areas, affluent communities today. And therefore AI for Social Good, as relevant to the primary needs of the developing world is what we call AI for Social Impact. So just to be clear, AI for Social Impact considers improvements in life and economic conditions for the underserved to be a first and foremost priority, above all other considerations.

3:50 Just to remind you what the UN Sustainable Development Goals are. These reflect problems across the developing world mainly, but also the West. There are 17 Sustainable Development Goals that were adopted by UN member states in 2015. And they are part of the 2030 Agenda for Sustainable Development. I'm not going to talk about each goal individually, but broadly, they can be classified into goals that impact society, that impact the economy, and that impact the environment. Now, in spite of these goals being adopted in 2015, there's been very limited progress, partly due to COVID, and most SDGs are not on track to be reached by 2030. And in fact, developing country targets on the SDGs are much more far off than developed country targets. And examples include poverty, which has been, poverty reduction has been slowed down by both COVID and the Ukraine prices. Food productivity is low, especially among small scale farmers in low and middle income countries. Small farmer productivity is less than $15 a day. The worldwide average income is less than $2,000 per year. And this is even lower in low and middle income countries, and among women.

5:17 So there's various indicators: disease. There's limited progress against TB, against malaria. This progress was slow to begin with and it's been further derailed by the pandemic. There are 10 million TB cases worldwide, for example, 30% of them are from India, this situation is expected to worsen. We don't have the 2022 numbers yet but we expect them to be worse.

5:45 Now, how does AI come in? There's been a paper that's about four to five years old now that came out of the McKinsey Global Institute that maps AI technologies, different kinds of AI technologies, to the various SDGs. And what was found in 2018, is that there were just 160 deployed use cases of AI that directly impacted the SDGs. Now, even if we imagine that number has doubled today, there is a huge addressable space where AI can be used. There's tons of problems that involve data. Many of these problems are, in fact, low hanging fruits for the use of AI technology.

6:30 It's also been shown in a recent Nature paper, that the positive impacts of AI significantly outweigh negative impacts, especially as they relate to addressing the SDGs. And there are pathways to mitigating some of the negative impacts. So for example, the use of green compute, there's a lot of research being done in the area of green compute. Human in the loop to address some of the inaccuracies and biases of AI. Having very strict regulation and control around deploying AI. Better data methods, you know, preparation methods, bias detection methods, and so on. Better evaluations of AI models, upskilling of the population to allow for more diversity in the AI workforce. All of these areas are currently being worked on across the world. So it's, it's really hope that you know, as time progresses, as far as it relates to addressing social issues, the positive impacts of AI will outweigh, will further outweigh the negative effects.

7:40 So, this brings us to our work, which as I mentioned already consists of building AI based solutions for underserved communities in the Global South. Our current focus is on India. We are a nonprofit, we are

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independently funded. Our portfolio spans several projects in health and agriculture. Now we'll be talking about some of these.

8:06 The overall approach we take is very similar, in fact, to an applied AI department within a technology company, except of course, the domains are very different, the deployment pathways are very different, and the data is very different. So the teams that build our AI solutions consist of a number of different types of people, starting from domain experts who are, you know, depending on the solution we're working on, would be doctors, biologists, entomologists, climate experts, and so on. Then there's a machine learning slash AI modeling team, which defines the approach to the problem and then actually solves the problem using AI. There's a user research and design team whose charter is to understand the beneficiary, the end user behaviors and patterns, and reflect these in the user interface within which the AI will be deployed. Obviously, we have an engineering team to build software for AI deployment and to build data pipelines. And then we have representation from monitoring and evaluation, whose job is to measure outcomes and monitor whether benefits have accrued due to deployment of the AI. So this is sort of the fairly standard thing that you would do in order to deploy AI models and this is reflected within the work that we do as well.

9:39 Now there's a number of challenges, some of which are really generic to AI and some of which are unique to the social impact space. So, these challenges can be bucketed in several categories. The first is the problem definition challenge, which consists of mapping the societal problem, which often is well understood to the AI problem, which often is much less well understood. The context doesn't always clarify the nature of the AI problem. Do you want to solve a classification problem? Do you want to solve a regression problem? Is it some type of reinforcement learning problem? What exactly are you trying to do mathematically, in order to address the societal problem? And what makes most sense? Sometimes there are multiple things that may have impact, but what's the most impactful thing to do? So that's the third question really: what specific problem, when solved, will make a greater difference in terms of impact? So this is somewhat the problem definition challenge, which is important to address right at the beginning.

10:48 Then there's the data challenge. Data in the social sector is far less clean than data in the technology sector. Data is noisy, it's incomplete, sometimes stale. It's very hard to refresh your data in many situations. It comes in sort of sporadically with non uniform velocity. There are obvious privacy challenges, because your solution is going to be directly used on people, people who don't necessarily understand technology very well. And you have to be very sensitive about this. Labels, annotations on the data, are also often noisy, especially when you use data that's been made for a very different context, and is now being used for building AI. In many cases, the labels require expert labelers, people who have qualifications, and because of that you can't label at scale, right, because you have access typically to very few, expert labelers.

11:54 You almost always don't have big data. It's small data. It's data that's often not completely representative of the community in which you will deploy. And you'll have to address that in various ways, either by collecting additional data, or evaluating really well, and making algorithmic adjustments, in order to make sure your model works well in the setting that you will deploy. And then integration of datasets. Often you want to use different datasets, you want to extract features from different datasets to use in the AI but the primary keys for these datasets are not very easy to join. I mean, the datasets are not easy to join through a common set of keys, even though they should be, so integration becomes difficult.

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12:45 So that's the Data Challenge. Then there's the metrics challenge, as you probably know, it's very important in AI problems to define the right metrics. That's not always clear. And in fact, this, you could consider this part of the problem definition challenge but it's important enough that it deserves its own sort of category. So one needs to define the right metrics to optimize for the AI that will eventually map to impact metrics or business metrics. And this mapping should be reasonably clear. And then you have to specify the right values for these metrics that the AI will need to reach in order to be acceptable or in order to have desired impact, or sometimes just even in order to be adopted, you know, even before it starts having impact. And this is often a very tricky problem, which you can't specify upfront. And the right values of these metrics is sort of an evolving target as you engage with the communities in which the AI will be deployed, this becomes clearer.

14:00 Then there's the adoption challenge, you know, who will adopt your AI? What is the cost in terms of money, time, labor and effort that's incurred by partners, which are typically NGOs or the government in adopting the solution and making sure that it sticks with beneficiaries that this, you know, the level of strap training that's involved to use the AI? Is there a willingness to adopt a black box algorithm that most machine learning algorithms are?

14:30 Then the deployment challenge. What's the tech stack in which the solution will be deployed? How will this integrate into existing public workflows? Do you need to create a completely new workflow in order to use the AI? These are all important questions. Once you deploy, there's the evaluation channel challenge. Typically, when you deploy AI solutions, you want to have some type of periodic or continuous evaluation process through on the ground measurements. This is often not very easy to do in the social context. You also want to measure differential impact of the AI versus not having the AI, and so you typically want to do AB testing. And this also can be challenging as I'll highlight in one of the problems that we're working on.

15:24 Finally, once you deploy, once you evaluate, there's the iteration challenge, where you have to bring data back in order to iterate on the model, retrain models, refresh models, and so on. And this is not just technical work, but it requires understanding and buy-in from partners on the necessity of iteration. Letting them know that an AI model is not a typical software system that you know, all in typical software systems, versioning in typically involves adding features, removing bugs, and so on. Whereas, really versioning in the sense of AI is a very fundamental thing, since the underlying data distributions can change. So in order to do that, you also want to be able to sample data from the deployment and continuously annotate it to enable retraining of these models. So this is sort of the whole or at least most of the spectrum of challenges that face you when you're doing AI in this space.

16:31 So with that, I want to get into some of the specific solutions we are working on, and maybe mention how these challenges play out in the solutions. So the first one I would like to talk about is automated pest management in cotton farms. I've listed here the team that worked on this problem. As you can see fairly large teams because of the nature of these problems - they require all kinds of expertise.

17:02 So some context behind this. Cotton happens to be one of the largest cash crops in the world. And about 90% of it is grown by smallholder farmers. These are farmers who own less than an acre of land. They struggle the most with uncertainty in yield and income. They're often driven to indebtedness in extreme cases. And in fact, extreme cases are not that uncommon. They're also driven to suicide. India happens to be the world's largest

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producer of cotton. It grows 26% of the world's cotton, about 40% of the farmland across the country is used to grow cotton. There are 6 million cotton farmers, about eight times as many people engaged in ancillary activities related to cotton farming. And most of these farmers, most of these cotton farmers, are smallholder or landless farmers. To top it all, cotton is particularly vulnerable to pests, right? And so, this is what really makes for a lethal combination of things.

18:15 The average annual cotton yield in India is less than 500 kg per hectare, which is significantly lower than the global average of over 750 kg per hectare. And one of the main reasons for this is that about 30% of crop yield is lost due to pest attacks. In a typical pest attack, approximately 70% of a farm is destroyed. And these pests are bollworms. They are a pest that attack cotton crops. And as a result of all this, you know you have smallholder farmers, you have pest attacks. The farmers are therefore extremely wary of pest attacks, and therefore tend to spray pesticide indiscriminately, which is bad in many ways, you know. It's bad for the environment, it increases farmer costs and so on. But the result of this is that 50% of India's total pesticide usage is in cotton.

19:20 So what we've developed in order to help farmers spray pesticide more judiciously, we've developed an AI based early pest warning and advisory system called CottonAce, which is an app that sits on a smartphone. The idea behind this is to enable farmers to protect their crops before the crops are damaged, right. And it also helps extension program officers who work with the farmers to monitor pest attacks. So CottonAce is really a set of 3 different apps - one for each type of end user. There's an app for cotton farmers, there's an app for extension officers, and then there's sort of a dashboard that aggregates data and can be used by agricultural program administrators.

20:14 But at the heart of it, the way this app works is that there's a demo farm in a village, typically one demo farm per village, where a pheromone trap is installed that attracts these pests. The cotton farmer empties the pest caught in the pheromone trap onto a white piece of paper, right, and just sort of spreads them out on the white paper, then they use our app to take two pictures of the trap, of the catch from the draft, not really the trap itself, but just the catch from the trap. And then the AI model in the back end counts the number of pests of each type in the trap. And it's learned to distinguish pests of different types, it's learned to distinguish pests from non pests, it's learned to distinguish real pictures of pests emptied on a web page versus other pictures and so on. And then based on that, based on the pest count, it then checks whether the pest count on the trap exceeds a certain threshold. If it does exceed a certain threshold, there is a spray recommendation, and the app recommends for the farmer to spray within a certain amount of time. And also tells you what the pest is. And any generates an advisory based on, you know, thresholds. So this is all in real time, the AI model is integrated into the phone, you know, this doesn't need cloud connectivity, because often the internet is not so great in these farms. So this is fully integrated into the phone.

22:04 Now once this advisory is given to the farmer at the demo form, or a lead form, as it's called, then the lead farmer sort of disseminates the advisory at village level to all the farmers in the village. And that's how we scale up the impact. In terms of the machine learning model, you know, there's an input image to a ResNet based network. And then there's two modules, one module that does image validation checks that this is a genuine image of pests on a piece of paper, the other module actually does the counting. And then these two work in tandem to result in an action recommendation. This sort of just getting into more detail around that.

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22:59 And then these are some of the results that we have. In the left panel, the Y-axis here is the mean absolute error in the pest count and then the X-axis is the ground truth value of the pest count. And you can see, this is a fairly hard problem in the sense that you can get high errors in the count, in the actual count. You can get errors of 10 pests or so. The blue and orange lines represent two different models. On the other hand, in terms of the accuracy of the recommendation, which is the right panel here, we consistently find that, you know, this orange model, for example, has an accuracy of over 80% in terms of the final recommendation. So there's quite some room for improvement, but on the other hand, the recommendation accuracy is fairly good.

23:54 So what are the challenges we faced here? There's a number of challenges in developing this. We had to create unique datasets and get them annotated by entomologists, people who can recognize pests and not just live pests, you know. There's lots of pictures of live pests in textbooks, but these are actually dead pests. The out of distribution problem is very challenging. You know, figuring out whether something is really whether an image really represents pests on a piece of paper. And this had to be addressed by a combination of modeling and good user interface.

24:33 The counting problem is exacerbated by data that is very non structured. So, for example, dead pests have body parts all over the place. The way the pests are emptied on the piece of paper, often they're clumped together and then it's very hard to count things like that. Installing traps in farms is a non-trivial process for the farmer, it can be expensive. Actually carrying a blank piece of paper that remains clean into the farm is a non-trivial process. Farmers don't often monitor the frequency, at the desired frequency. So we are working on some trap free approaches where we use remote sensing data to predict pest infestations as well. And the vision is that we would use both these models in tandem.

25:24 Current approaches are not easily scalable to multiple pests and crops, and there we're looking at synthetic data generation techniques, especially in the light of recent advances with stable diffusion, and so on. And we're looking into this. There's a lot of farmer skepticism behind using yet another app. There exists a lot of agriculture based apps for farmers, and pushing adoption through government partnerships. So that's what we're really trying to do is to set up government partnerships in various states of India, and then work with the government to get adoption through farmers. So these are some of the challenges we faced here. But we're slowly deploying this solution, and it's growing year by year in terms of the number of farmers who are adopting it. We've also open sourced all the data behind this. And in fact, we had a bollworm counting competition with that data. And we are just putting together, summarizing the results of this competition, and these results will be available for the community on our website.

26:34 The next project I wanted to talk about was the newborn anthropometry project. Newborn babies, sort of risk to life in newborn babies, it's well known that the first 42 days of life are the most vulnerable time for a child's survival. And children face the highest risk of death in this period. And it turns out that infant mortality, particularly in these days in its first six weeks, first month even, is strongly correlated with their weight. In India, in particular, three neonates die every minute. Every fourth baby that's born is low birth weight, right. And there's a number of statistics around low birth weight and infant mortality that I won't get into in detail.

27:32 What we're doing is a smartphone based technology, again, that will help frontline health workers identify underweight babies and to monitor their growth. And once you can do that in a systematic way, then you can drive

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interventions in the communities where this is used. So the idea is really to use this in rural communities, where weight is not accurately measured, because weight is typically measured with spring balances. These spring balances have errors. There's errors that are intrinsic to the spring balance, you know, they're not well calibrated, and so on. And there are errors in recording the data that they use from the spring balance. And what we aim to do is sort of eliminate all of these errors by having a simple smartphone based app, which you can use to take a picture or a video of a baby and log the weight in the back end.

28:33 So, the overview is that health workers take a video of the newborn in a home setting using our app. The AI model estimates the newborn's weight and other anthropometric measurements like head circumference, chest circumference, and so on from the video. The worker makes recommendations based on weight and other danger signs, and the newborn gets appropriate care. In addition, databases at the back end that may influence policymaking are updated automatically. And so health systems get accurate real time, tamper proof evidence, tamper proof prevalence of low birth weight by region, right. So you're both sort of driving interventions for individual families, as well as have good, reliable aggregate numbers for low birth weights across communities. So the idea is to use this and replace spring balances and to enable tamper proof and automatic digitization. So it's really a two pronged thing. It's not just accurate weight estimation, but also the digitization process.

29:55 So the way this works is, we call this app Digital Taraju, and there's some information that's entered by the healthcare worker prior to taking a video of the baby. There's a wooden ruler that's put next to the baby in order to enable the AI to estimate depth. And I'll come back to this, we actually experimented with a number of different reference objects, including the wooden ruler. But it turned out that the wooden ruler was the simplest thing to use. And then, since currently this is a data collection app and also enables ground truth data collection, the worker, the health worker, called an ASHA worker, also measures the length of the baby, the chest circumference, checks whether the baby has a temperature and so on. In fact, they also measure the weight of the baby using a spring balance and enter it. And then you take the picture, and then there's some advisory that's generated based on the weight of the baby.

31:03 In terms of the technical approach for doing this, we've explored a number of different approaches. We started with what's called a 3D model based approach. Where we use the data to create 3D models, to create a 3D model of the baby. And so the AI is used to create a 3D model. You measure the volume inside the 3D model in order to estimate the weight. So this is one way to do it. The other way is a more direct method, which is called the model free method, where you extract the frames from the video, and directly regress to the weight and other anthropometric methods, other anthropometric quantities without having an intervening 3D model of the baby. It turns out with the data that we have, it makes more sense to use this more direct approach. The 3D modeling approaches are far more data hungry than the direct model free approaches.

32:08 Now within these model free approaches, we focus primarily on two directions. One is multi-task learning, where the model learns the baby weight and other parameters by predicting related tasks. And the other is contrastive learning where the model learns through video comparisons of different babies, not just at the individual baby level. So in the first type of multi-task learning, we don't just predict anthropometric parameters, but we also predict a segmentation mask that isolates the baby. It turns out, if you do this jointly with the anthropometric parameters, you get higher accuracy for the anthropometric parameters. Similarly, you can also try and predict important key points on the body of the baby and that also enhances the accuracy of predicting other parameters. In contrastive learning, we actually force features that are collected from features that are computed from babies of similar weight to be closer together in feature space, then features that are collected, or that are computed from babies of significantly different weights. And this way we sort of aim to improve accuracy of the metric. So we're using both of these techniques currently.

33:35 And just some results here. So first, in a hospital setting, we do really well, because the lighting is uniform. You know, the backgrounds are uniform, all the babies are placed on white sheet. And in those kinds of control environment settings, our results are really good. I mean, if you, for example, look at the two class problem where you are predicting whether a baby has weighed less than two and a half kg versus more. Both the recall and precision are very high. Even the weight estimation problem in a controlled setting, we do really well. In fact, our average error in weight estimation is 75 grams, and 50% of babies have error less than 50 grams, 90% less than 150 grams. So we do really well in a controlled environment.

34:30 It turns out to be not so easy in a home environment where backgrounds can vary and where lighting conditions are often poor. So we are in the process of doing some of these experiments. What we clearly find is that the error goes down with the number of training samples, so we haven't yet converged. We continue collecting data. I have already described some of the modeling methods here. And the best result we get so far in the home setting is an average error of 190 grams. And in fact, we're working on improving this further by adding other tabular data referring to the environment in which the baby is.

35:20 So we're fairly confident we can solve this problem to a level that this is usable. But there's various challenges that we had to go through to get here. First of all, the problem was somewhat ill defined from the public health end in terms of whether we wanted to really estimate weight, which is a harder problem. No sorry, really estimate whether the baby's low weight or not, which is an easier problem, versus monitoring the growth with time. So to monitor growth with time, you have to estimate the true weight at various time points. Whereas if you only want to tell whether the baby has low weight or not, it becomes a classification problem, which is easier. We've finally converged with extensive, you know, after extensive discussions with domain experts, with public health experts, that growth monitoring is the right problem to solve. We had no sense of a good target metric for this problem, because the cumulative error of the current manual system is very difficult to estimate. And so we were working blindly in terms of the accuracy that the AI needs to get to in order to be usable. We're trying to remedy that currently by doing a study. We are about to start a study where we actually measure the cumulative error of the current manual system.

36:49 There are huge data challenges here. Data collection challenges. There’s variability of backgrounds and lighting, as I've mentioned. There are very strong deployment constraints around the type of phone that is usable. We have to make sure we collect all our data with phones that will actually be used in the field by health workers. And also, we had constraints around the type of reference objects that are usable for depth estimation. So for example, a reference object that works really well in this case, is a checkerboard. But checkerboards are not objects that can be scaled in the field. They cannot be assigned easily to health workers staff, and therefore we converge on the use of a wooden ruler.

37:33 We have to train staff to take these videos in very difficult work environments. Internet availability is unclear. Currently, we're still working with models that are large and will therefore have to be deployed on the

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cloud but at some point, we are going to have to compress them to fit on a basic smartphone, and that's going to affect accuracy. Pulling in contextual data related to the location is also going to be a challenge, because it will require cloud connectivity. And in general, anything to do with babies and mothers requires a lot of creation of trust and strengthening of systems. So this is a fairly challenging problem for these reasons.

38:15 The last problem I'd like to discuss here is our prediction model for loss to follow up in TB patients. And again, some context here. TB, not COVID is the world's leading infectious disease. India accounts for about 26% of the global burden. There's two and a half million new TB cases in India every year. There's half a million deaths due to TB every year, just in India, and three times as many worldwide, so one and a half million worldwide. And the number of total reported cases across the country is about 2 million. So it's a huge, huge problem.

39:08 TB is a curable disease. But one of the issues with TB is loss to follow up. Now, loss to follow up is an extreme form of non-adherence to the medication regimen, where the treatment is interrupted for one or more consecutive months. So just to give you some context here, it turns out that if you miss even 10% of your medication doses for TB, there's a very high chance that you get drug resistant TB, which is very difficult to cure. And this is sort of the leading cause for TB related deaths.

39:46 About 100,000 patients per year across the country were deemed loss to follow up because they just stopped taking the medication for one month or more. And so what we'd like to do is really to identify in advance if a certain patient is at risk of loss to follow up so that TB workers can take appropriate interventions. They can go and visit the patients homes, persuade them to take their drugs regularly, and so on. Now, these are very intensive interventions, so it's important to know in advance which patients are likely to need these interventions, and that's where the AI comes in.

40:31 So the way this solution works, the way this AI solution works, is that the Indian government has a large database of TB patients. This is longitudinal data. This database is called NI-KSHAY. And that is combined with other adherence related datasets to predict whether, to predict in advance whether, a certain patient is going to be loss to follow up within six months of treatment. So the machine learning pipeline is fairly standard. We got all of this data. We did some EDA on top of it, cleaning imputation. The data was very noisy, in fact, and then experimented with not just algorithms, but different types of encodings and data augmentation techniques. We evaluated this on a forward split, where we split patients in time rather than randomly, carried out modeling and hyper parameter optimization and then evaluated this.

41:43 So some results. In fact, we find that a lot of methods do equally well. The metric here that's important is we call it "k". So if you've got if you've got 100 patients in the community, and you can only, you only have enough health worker resource to target, say, k percent of these 100 patients, then the right question to ask here is that when you target k percent of the patients, what is your LFU recall? You know, what, what percentage of, what proportion of, LFU patients are you able to capture within the top k percent? And if it were, if this were a random model, you'd be following this black line here, which is that, you know, the recall at k would be exactly k. So any left above that black line is a positive lift. And in fact, these curves you see here, with various methods that we use are pretty much sitting on top of each other. We've tried various kinds of encodings, here, various kinds of models.

And we've also built a local explanations framework for figuring out what contributes the highest for each patient towards their prediction.

43:12 So I think I've already described this, I'm going to skip the machine learning overview. So there's again, various challenges to doing this. We have a lot of data for a project like this, because of the number of TB patients across the country and the fact that every patient is captured in this longitudinal database. But the data is extremely noisy. The features are noisy, the labels are noisy, patients are not often accurately labeled as LFU, there is missing data. So we had to work very closely with government experts to clean this up.

43:50 And turns out, it's very hard to improve models further. With algorithmic tweaks or feature engineering, I think the bigger lifts will probably come only from having better data and more data. The AI risk score, which is a number between zero and one, had to be converted to a binary decision to select patients who will receive intensive counseling. And this turns out to be a tricky problem, because there's huge variability in health worker resources across locations. So we have to make this something that is tunable, this parameter, something that is tunable at a very local level, and tunable by public health experts. And we're still trying to figure out what the best way is to do that, and to make sure that the solution is still accepted and adopted.

44:42 Typically in tabular datasets, there's huge potential for bias. So we had to test this extensively at a cohort level. And we're only comfortable piloting because we've convinced ourselves that the accuracy is reasonably uniform across cohorts that matter. The potential impact in this type of problem is large, but it's long timescale because it takes six months to go through TB treatment. And you will only know whether your solution has been successful after patients actually complete treatment and recover. So the on-the-ground evaluation of the AI is very, very challenging. There's no short time feedback loops. There's some ethical concerns behind doing AB testing, because then you'd have to deny a certain set of patients the advantage of the AI solution, and the only way you can reconcile that is that in fact, today, they don't have the advantage of that solution anyway. In general, anything that gives a score to a citizen carries inherent risks. And we need to be very careful about that, about releasing these scores, managing them well, periodically purging them as necessary, retraining models to make sure the scores are accurate and so on.

46:00 So I'd like to end there with an acknowledgement to all our funding agencies, as well as our government partners. Thank You.

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